

P. BHUVANESH

Robotics Engineer | ROS2 & Autonomous Systems Developer

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GitHub: github.com/Bhu470/dashboard | Portfolio: bhu470.github.io/dashboard/

PROFESSIONAL SUMMARY

Robotics Engineer with experience in ROS2, Gazebo, RViz, URDF/XACRO, autonomous mobile robots, and embedded robotics systems. Skilled in robot simulation, TF validation, sensor integration, SLAM concepts, and autonomous navigation workflows using Linux/Ubuntu environments.

EXPERIENCE

Robotics Engineer | Evolve Robotics Lab, Chennai | Mar 2025 - Present

- Developing ROS2-based robotic software modules, Gazebo simulations, and URDF/XACRO workflows.
- Working on autonomous navigation, sensor integration, TF validation, and robotics debugging.

Robotics Engineer Intern | Evolve Robotics Lab, Chennai | Oct 2024 - Mar 2025

- Developed ROS2 nodes, Gazebo simulation workflows, and robot visualization pipelines.
- Worked on SLAM testing, obstacle avoidance, and embedded robotics integration.

TECHNICAL SKILLS

Robotics & Autonomous Systems: ROS2 Humble, Gazebo, RViz, URDF/XACRO, TF2, SLAM concepts, autonomous navigation, differential-drive mobile robots

Programming & Platforms: Python, C++, Embedded C, Linux/Ubuntu, GitHub

Embedded Systems: ESP32, Arduino Uno, AVR, PIC, MSP430, encoders, ultrasonic sensors, L298N motor drivers

Communication Protocols: UART, SPI, I2C

AI-Assisted Development: OpenAI Codex, Claude Code, Cursor, OpenCode (learning and prototyping)

PROJECTS

ROS2 Differential Drive Autonomous Mobile Robot - Developed a ROS2 mobile robot using URDF/XACRO, Gazebo, RViz, TF validation, and encoder integration.

Autonomous Obstacle Avoidance Robot - Built an autonomous robot using ultrasonic sensors, Arduino, and differential-drive control.

Encoder-Based Mobile Robot Monitoring System - Developed encoder-feedback monitoring and synchronized motor-control workflows.

Robot Visualization & Simulation Workflow - Configured Gazebo and RViz for robot simulation, visualization, TF inspection, and sensor-data validation.

EDUCATION

B.E. - Electrical and Electronics Engineering | Sriram Engineering College | 2018 - 2022

Diploma - Electrical and Electronics Engineering | Sriram Polytechnic College | 2016 - 2018

CERTIFICATIONS

Robotics Internship - Evolve Robotics Lab | Embedded Systems - BYTESINBITS | MSP430 Programming - Udemy | IoT Training - BSNL